

# Daily Review

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Booklet p.10-11 — Getting Around in Space

- 1 Engine off in deep space. Describe the motion.
- 2 Describe the catapult manoeuvre.
- 3 How does an ion drive reach a high speed?
- 4 Why is there no friction in deep space?
- 5 What is thrust?

Constant speed, straight line

Gravity pulls it round, faster

A tiny force, for months

There is no air

The push from an engine

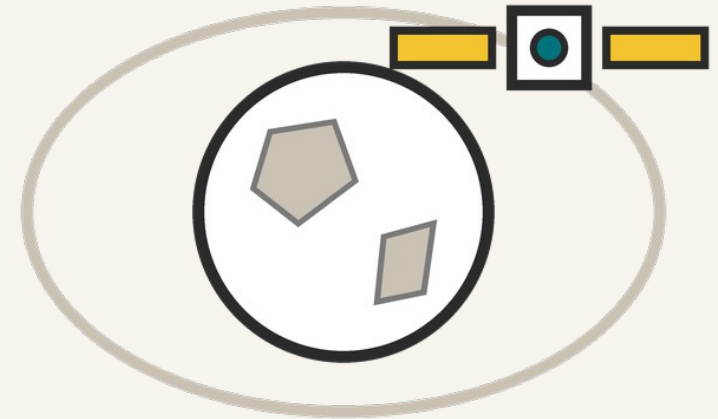
SPACE

# Satellites

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Weather, GPS and phone calls — all from objects falling round the Earth.

National 5 Physics (S3)



## THE BIG QUESTION

One satellite hangs still above this coast. Another crosses it in ninety minutes. What sets the difference?



Scotland from orbit — NASA

PART 1

# What satellites do

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What satellites do for us, and why they do not fall down.

## By the end of this part...

- 1 I can describe the benefits of satellites
- 2 I can explain how a satellite stays in orbit

# Show me — Describe each word

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SHOW ME BOARDS

1 satellite

An object in orbit  
around another

2 orbit

The path one object  
takes around another

3 period

The time for one  
complete orbit

4 altitude

The height above the  
surface

5 geostationary

Always above the same  
point on Earth

# Task 1 – what are satellites for?

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Booklet p.12

Write down as many uses and benefits  
of satellites as you can.

**Booklet p.12, Task 1**

# Task 1 – Answers

ANSWERS

Booklet p.12



**GPS**



**Weather forecasts**

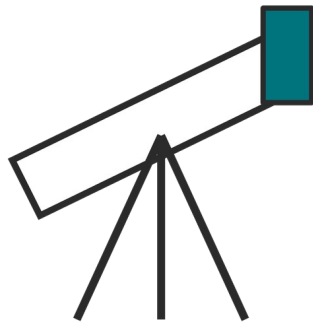


**Communications**

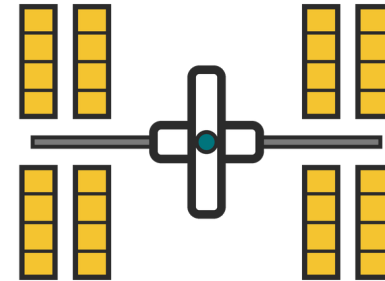
# Task 1 – Answers

ANSWERS

Booklet p.12



**Scientific discovery**

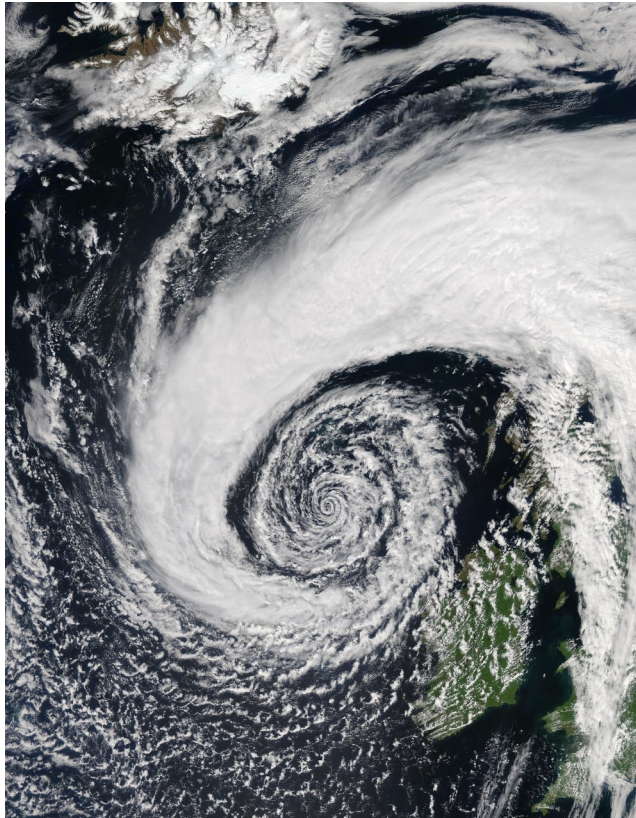


**Space exploration**

# Watching the weather

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A low pressure system off Ireland



- Satellites photograph the cloud from above.
- This spiral is a deep area of low pressure.
- Forecasters track it as it moves.
- Warnings can come days ahead.



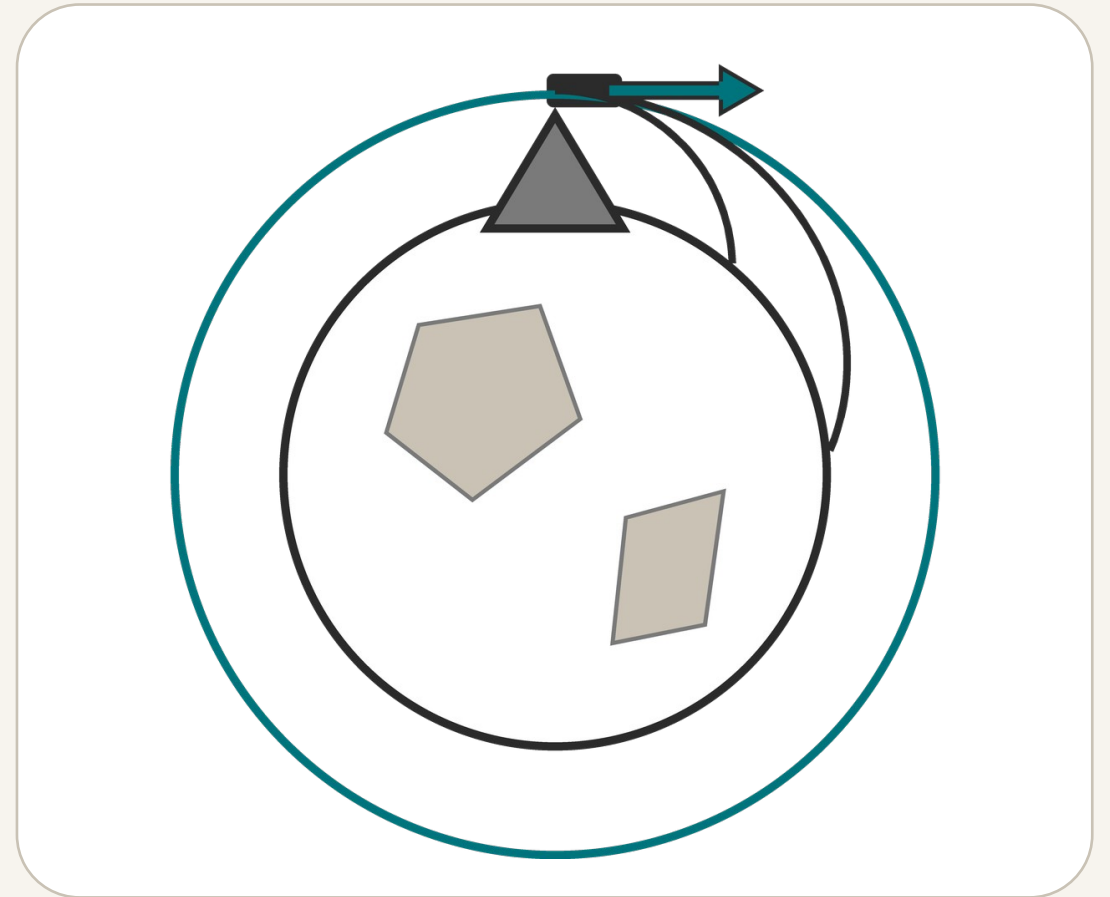
The Whirlpool galaxy — NASA / ESA Hubble Space Telescope

# Newton's cannon

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Booklet p.12, Task 2

- Fire a cannonball sideways from a very high mountain.
- The faster it goes, the further round it lands.
- Fast enough, the Earth curves away as fast as it falls: an orbit.



## Task 2 – Newton's cannon

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Booklet p.12

Watch the animation, complete the diagram,  
then describe what happens underneath.

**Booklet p.12, Task 2**

# Hinge question

FINGER VOTE

Think first — then show A to E on your fingers, 1 to 5

A satellite orbits 400 km above the Earth.  
Which statement about the satellite is correct?

- A** There is no gravity acting on it
- B** It is falling, and keeps missing the Earth
- C** Its engine keeps it up there
- D** It is beyond the Earth's gravity
- E** It moves in a straight line.

Answer B

# Show me — orbits

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SHOW ME BOARDS

EXTRA PRACTICE

- 1 What does “period” mean for a satellite?
- 2 What is meant by “altitude”?
- 3 Why does a satellite not crash into the Earth?
- 4 Name a satellite used for scientific discovery.
- 5 How do satellites help forecasters?

The time for one orbit

Its height above the surface

It falls as the Earth curves away

Hubble

They photograph the clouds

# Daily Review

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Booklet p.12 — Part 1

- 1 Name three uses of satellites.
- 2 Why does a satellite stay in orbit?
- 3 What does the period of a satellite mean?
- 4 What is altitude?
- 5 What is a satellite?

GPS, weather,  
communications

It falls as the Earth  
curves away

The time for one orbit

Height above the surface

An object orbiting  
another

## PART 2

# Orbits, altitude and period

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Choose the height and you have chosen the time for one orbit.

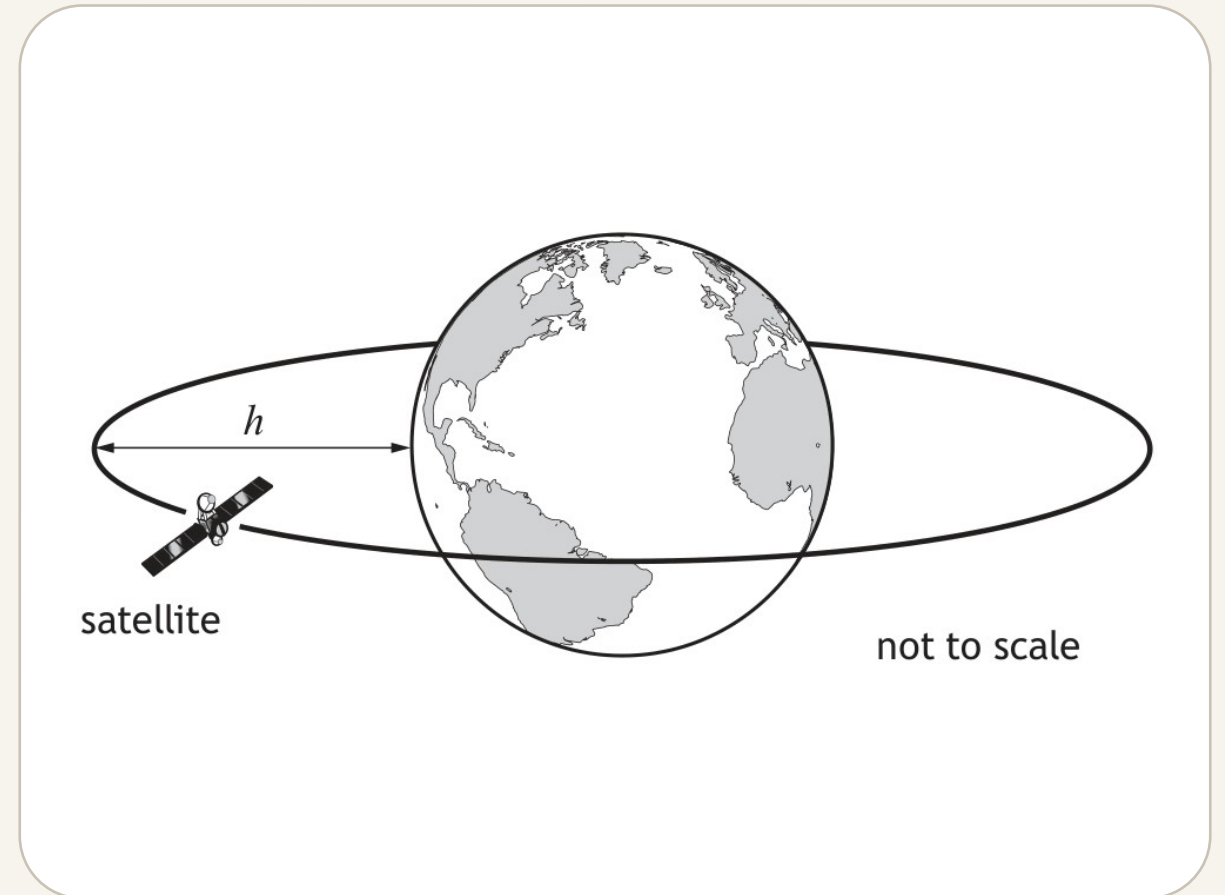
## By the end of this part...

- 1 I can state the altitude and period of a geostationary satellite
- 2 I can describe how altitude affects the period of a satellite

# Geostationary satellites

Booklet p.12, Task 3

- A period of 24 hours.
- An altitude of 36 000 km.
- It stays above one point on the Earth's surface.



# Task 3 – geostationary satellites

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Booklet p.12

Complete the definition.

**Booklet p.12, Task 3**

## Task 3 – Answers

### ANSWERS

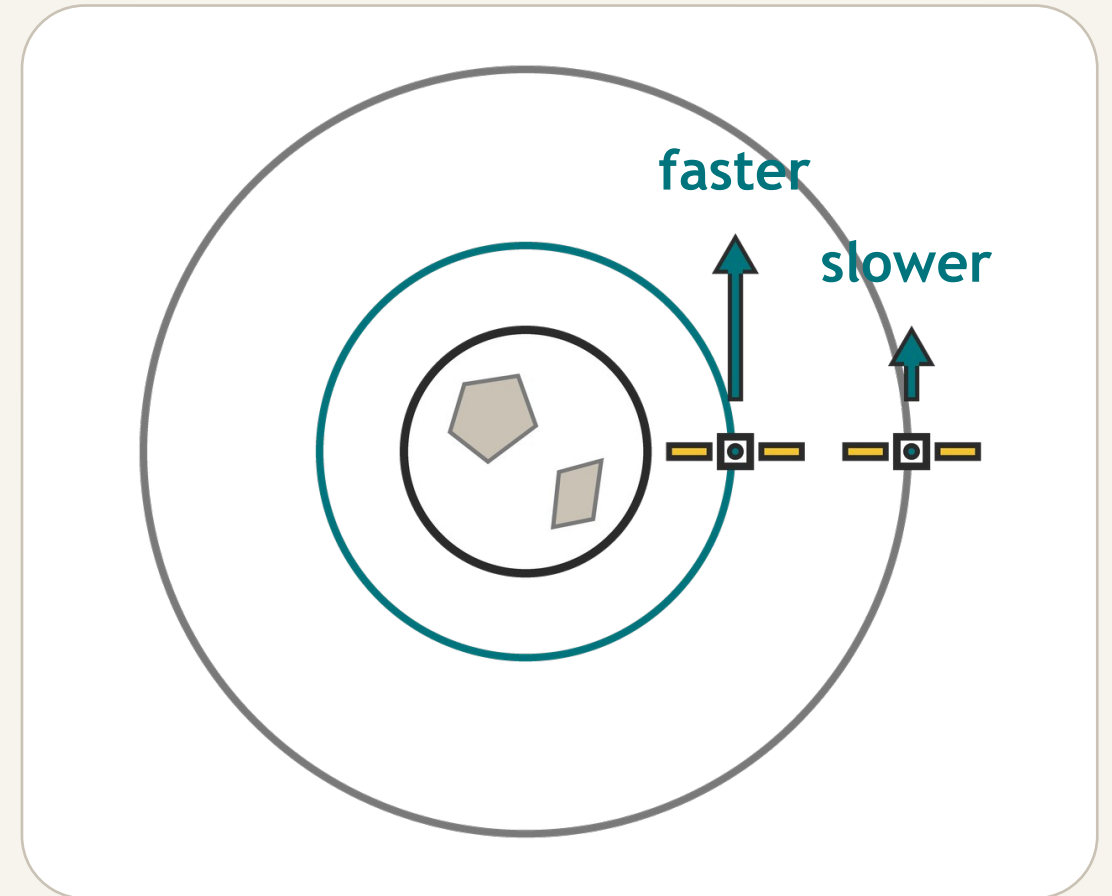
Booklet p.12

| Geostationary satellite |           |
|-------------------------|-----------|
| Orbital period          | 24 hours  |
| Orbital altitude        | 36 000 km |

# Why a higher orbit takes longer

Booklet p.12, Task 4

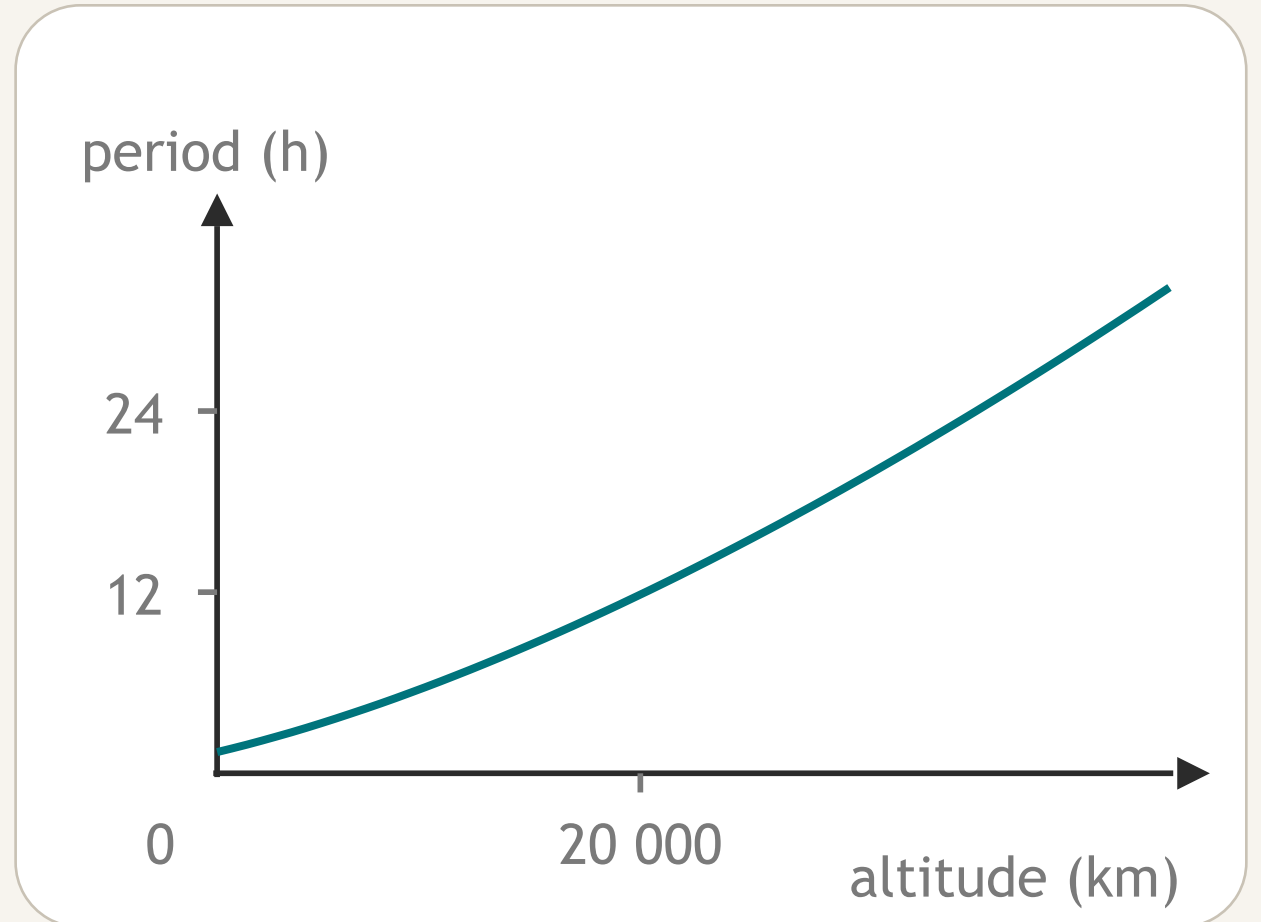
- A higher orbit is a bigger circle: further to travel.
- Gravity is weaker up there, so the satellite moves more slowly.
- Further to go, and going slower — so the period is longer.



# Higher orbit, longer period

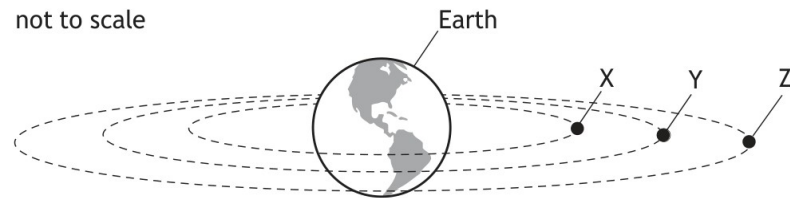
Booklet p.12, Task 4

- The higher the orbit, the longer the period.
- The ISS, about 400 km up: 93 minutes.
- Geostationary, 36 000 km up: 24 hours.



# Show me – X, Y and Z

SHOW ME BOARDS



1

Which satellite has the shortest period?

X

2

Which has the longest period?

Z

3

Z is geostationary. What is its period?

24 hours

4

Can X have a period of 24 hours? Why?

No: too low

# Task 4 – altitude and period

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Booklet p.12

Fill in the blanks.

**Booklet p.12, Task 4**

# Task 4 – Answers

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## ANSWERS

Booklet p.12

- As the orbital altitude of a satellite increases, the orbital period increases.

# Show me — predict

SHOW ME BOARDS

| Altitude (km) | Period      |
|---------------|-------------|
| 700           | 99 minutes  |
| 2000          | 127 minutes |
| 10 000        | 5.8 hours   |

1

Predict the period at 1000 km.

105 minutes

2

Is 20 000 km more than 5.8 hours?

Yes

3

111 minutes: what altitude?

1300 km

4

What pattern does it show?

Higher,  
longer

# Show me — predict

SHOW ME BOARDS

EXTRA PRACTICE

| Altitude (km) | Period      |
|---------------|-------------|
| 408           | 93 minutes  |
| 1000          | 105 minutes |
| 36 000        | 24 hours    |

1

Estimate the period at 600 km.

97 minutes

2

Is 20 000 km more or less than 24 h?

Less

3

Is the satellite at 408 km geostationary?

No: 93 min

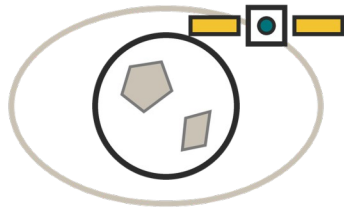
4

Which satellite goes round most often?

The 408 km one

# Scotland in space

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**UKube-1, Glasgow**



**SaxaVord, Shetland**

- UKube-1, Scotland's first satellite, was built in Glasgow (2014).
- SaxaVord on Unst, Shetland, is licensed as a UK spaceport.

# Fly the orbits yourself

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## TRY THESE

NASA Eyes on the Solar System: fly to any planet.  
PhET Gravity and Orbits: change the speed and watch.  
Try Newton's cannon: how fast is fast enough?

# Extension – BBC Bitesize

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Booklet p.13

Revise space exploration on  
BBC Bitesize.

Booklet p.13, Extension



BBC Bitesize

# Checkpoint

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- 1 Give three benefits of satellites.
- 2 State the period and altitude of a geostationary satellite.
- 3 Satellite A orbits higher than B. Which has the longer period?

GPS, weather,  
communications

24 hours; 36 000 km

A — higher means longer

# Exam practice

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## Past paper and specimen paper

- National 5 Physics 2023, Section 2, question 3(a). 2 marks.
- Question 3(c). 1 mark. Predicting from a satellite table.
- Specimen paper, Section 2, Q2(d)(ii). 2 marks. Why the ISS stays up.
- Specimen paper, Section 2, Q8(a). 3 marks. A GPS signal's distance.
- Then booklet p.14, Task 5.

LOOKING AHEAD

# Forces on Rockets

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Thrust, weight and Newton's laws — what it takes to leave the pad.

# Plenary

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- 1 The big question: why does one satellite hang still over Scotland?
- 2 Name three things satellites are used for.
- 3 What is the period of a geostationary satellite?
- 4 What is its altitude?
- 5 What happens to the period as the altitude increases?

It is geostationary: a 24 h period

GPS, weather, communications

24 hours

36 000 km

It increases

# Plenary

## EXTRA PRACTICE

- 1 What is meant by the period of a satellite?
- 2 Why does a geostationary satellite stay above one place?
- 3 Which has the shorter period – the ISS or a TV satellite?
- 4 Why does a satellite stay in orbit?
- 5 Describe what happens in Newton's cannon.

The time for one orbit

It goes round once a day

The ISS

It falls as the Earth  
curves away

Faster shots land further  
round

# Extension

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- 1 Why must a TV satellite be geostationary?
- 2 Suggest why weather satellites are often in low orbits.
- 3 Why does a satellite in a low orbit need to travel faster?

# Extension

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## EXTRA PRACTICE

- 1 Suggest why GPS needs many satellites, not just one.
- 2 Why can the ISS be seen crossing the sky in a few minutes?
- 3 Suggest a problem with 36 000 km for a phone call.

# Course specification

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## N5 Physics course specification v4.0, p.8 – Space exploration

- Awareness of the benefits of satellites: GPS, weather forecasting, communications, scientific discovery and space exploration (for example Hubble telescope, ISS).
- Knowledge that geostationary satellites have a period of 24 hours and orbit at an altitude of 36 000 km.
- Knowledge that the period of a satellite in a high altitude orbit is greater than the period of a satellite in a lower altitude orbit.